

LSTM vs Transformer

Stock Price Prediction — Fair Comparison Report

Fresh inference run: 2026-04-28 00:48

Symbols evaluated: 17 / 17 • Test split: last 15% (chronological)

Avg MAPE → LSTM: 6.27% | Transformer: 9.79%

Avg R² → LSTM: -0.963 | Transformer: -2.442

Avg DirAcc → LSTM: 50.5% | Transformer: 50.5%

Architecture — LSTM:

3 × LSTM layers (128 → 64 → 32 units) + BatchNorm + Dropout + Dense head

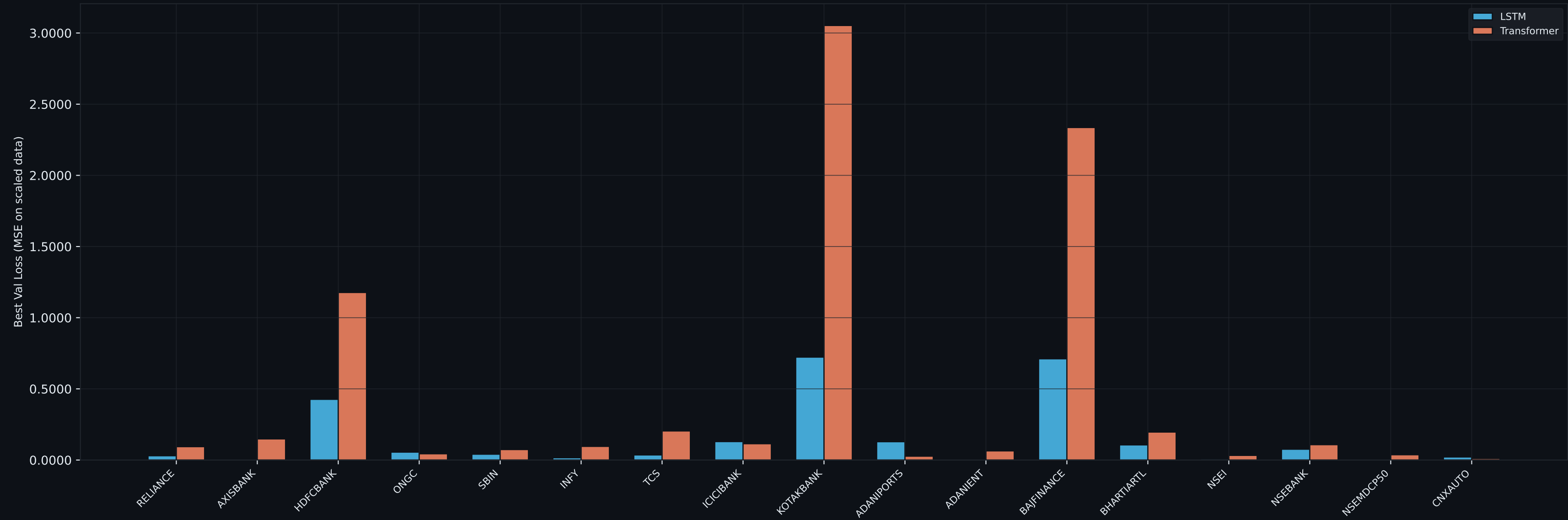
Architecture — Transformer:

3 × Encoder blocks (Multi-Head Attention 8h×64 + GELU FFN 256) + Positional Embedding

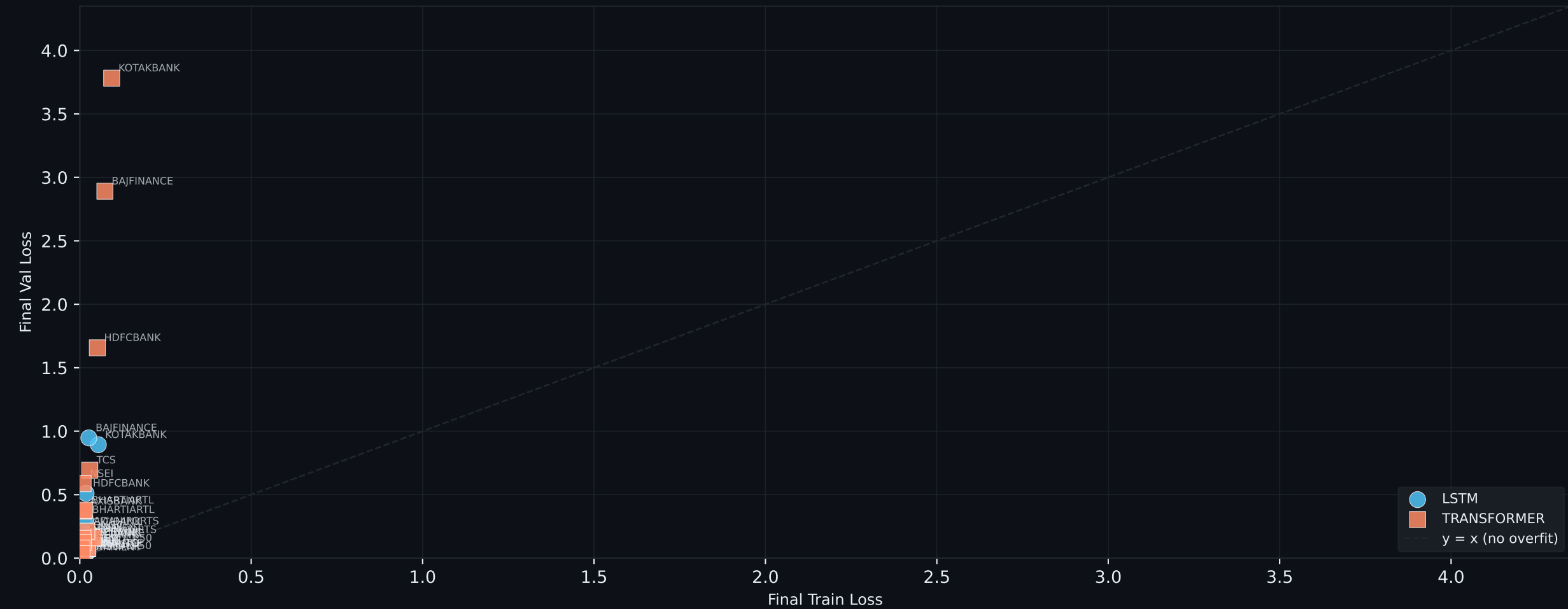
+ GlobalAveragePooling + MLP head (128 → 64 → 1)

Train/Val/Test: 70%/15%/15% chronological • Seq len: 120 • Seed: 42 • Period: 5 years

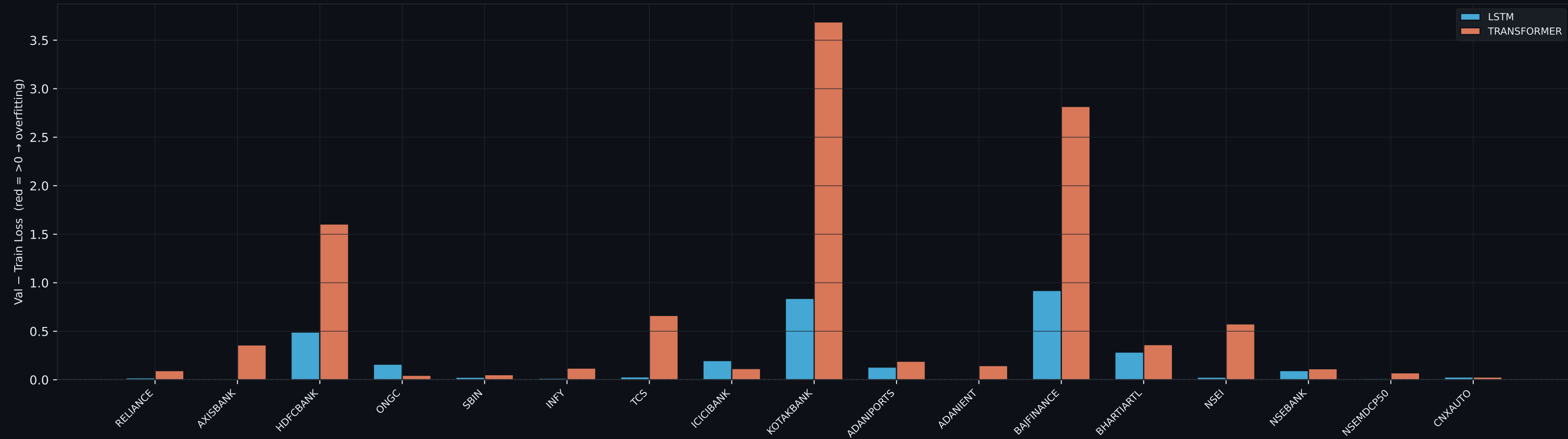
Training Summary — Best Validation Loss per Symbol



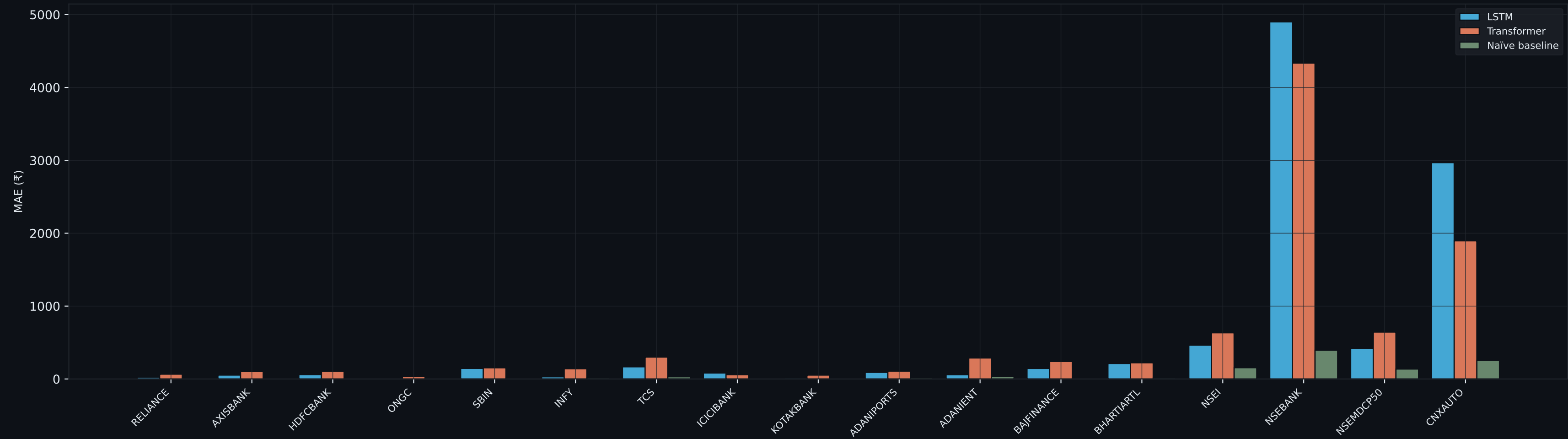
Overfitting Check — Final Train Loss vs Final Val Loss



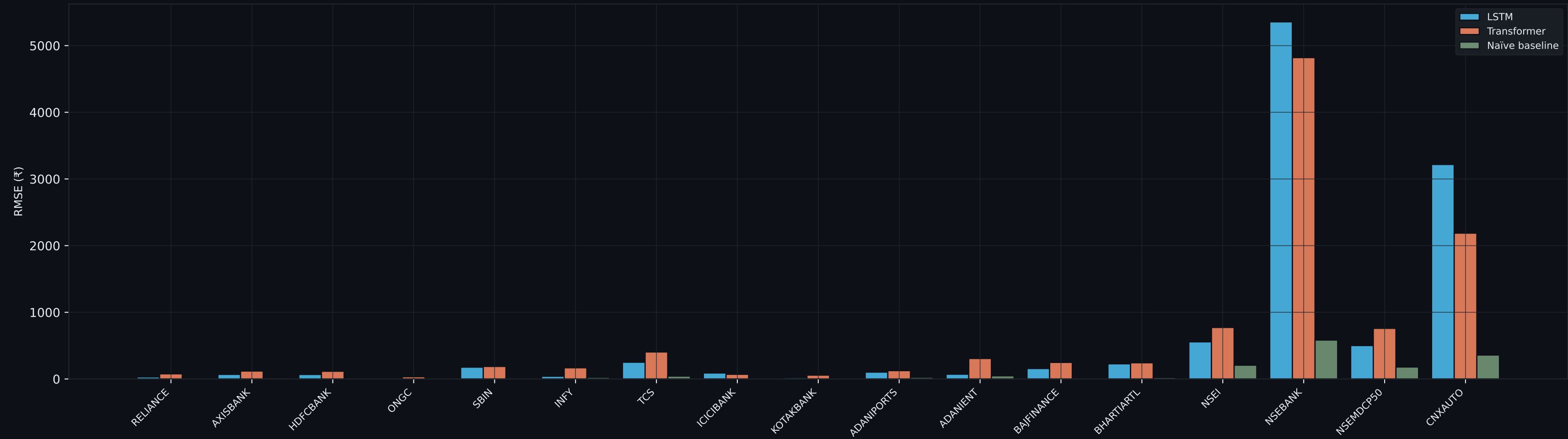
Train-Val Loss Gap (Val – Train) — Overfitting Indicator



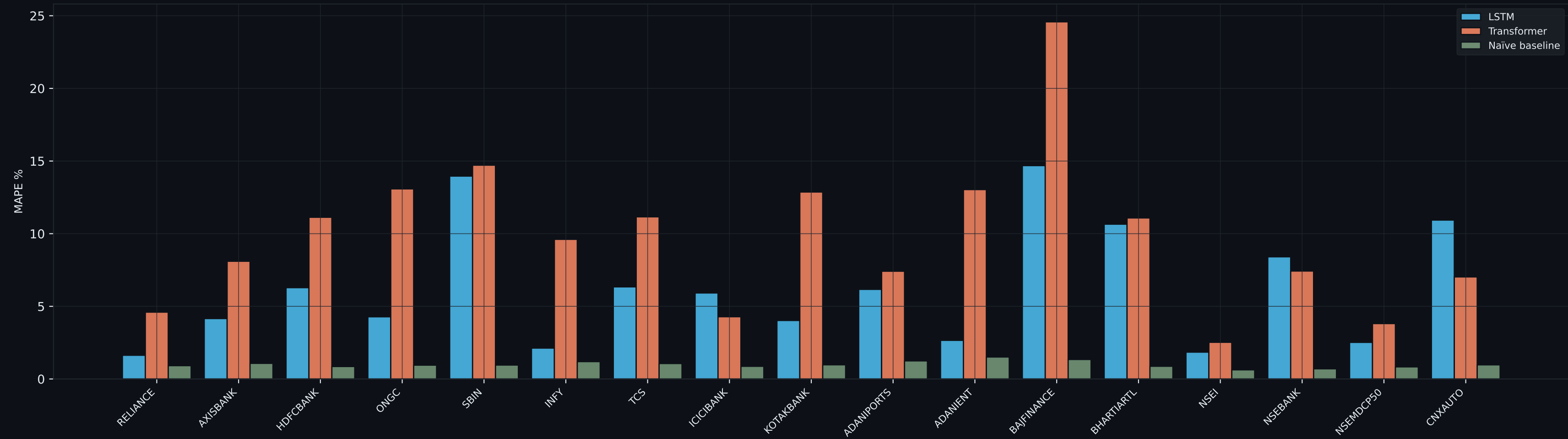
Model Comparison — MAE (₹)



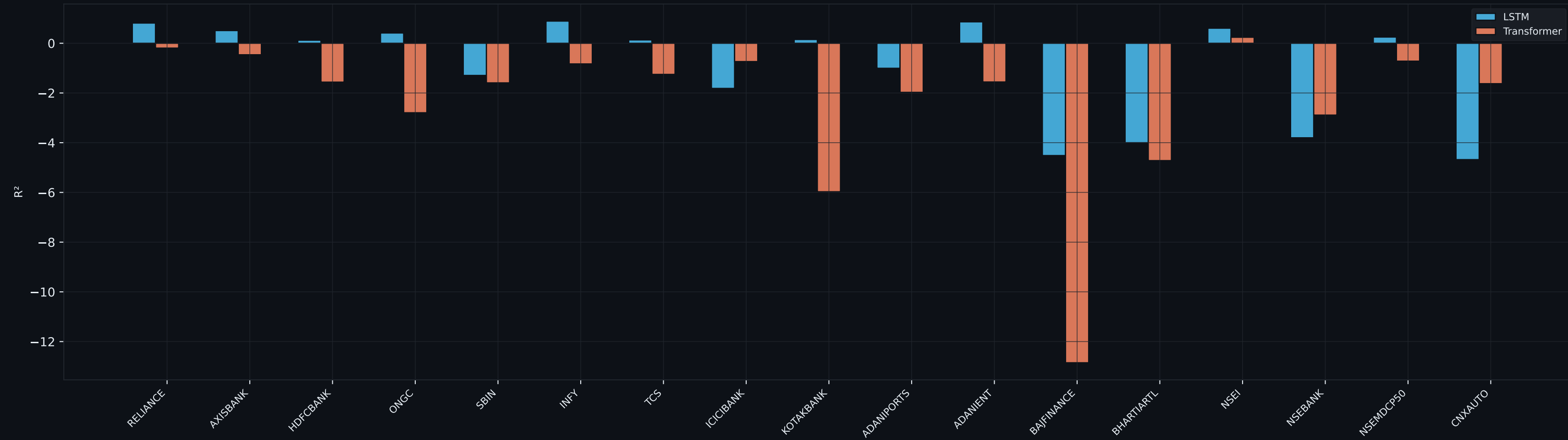
Model Comparison — RMSE (₹)



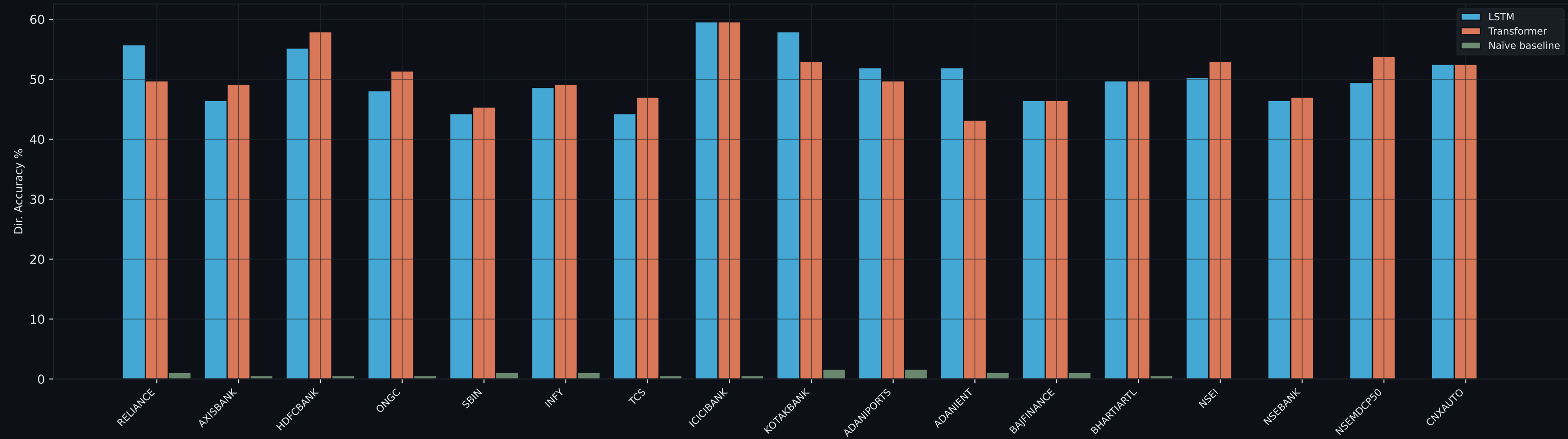
Model Comparison — MAPE %



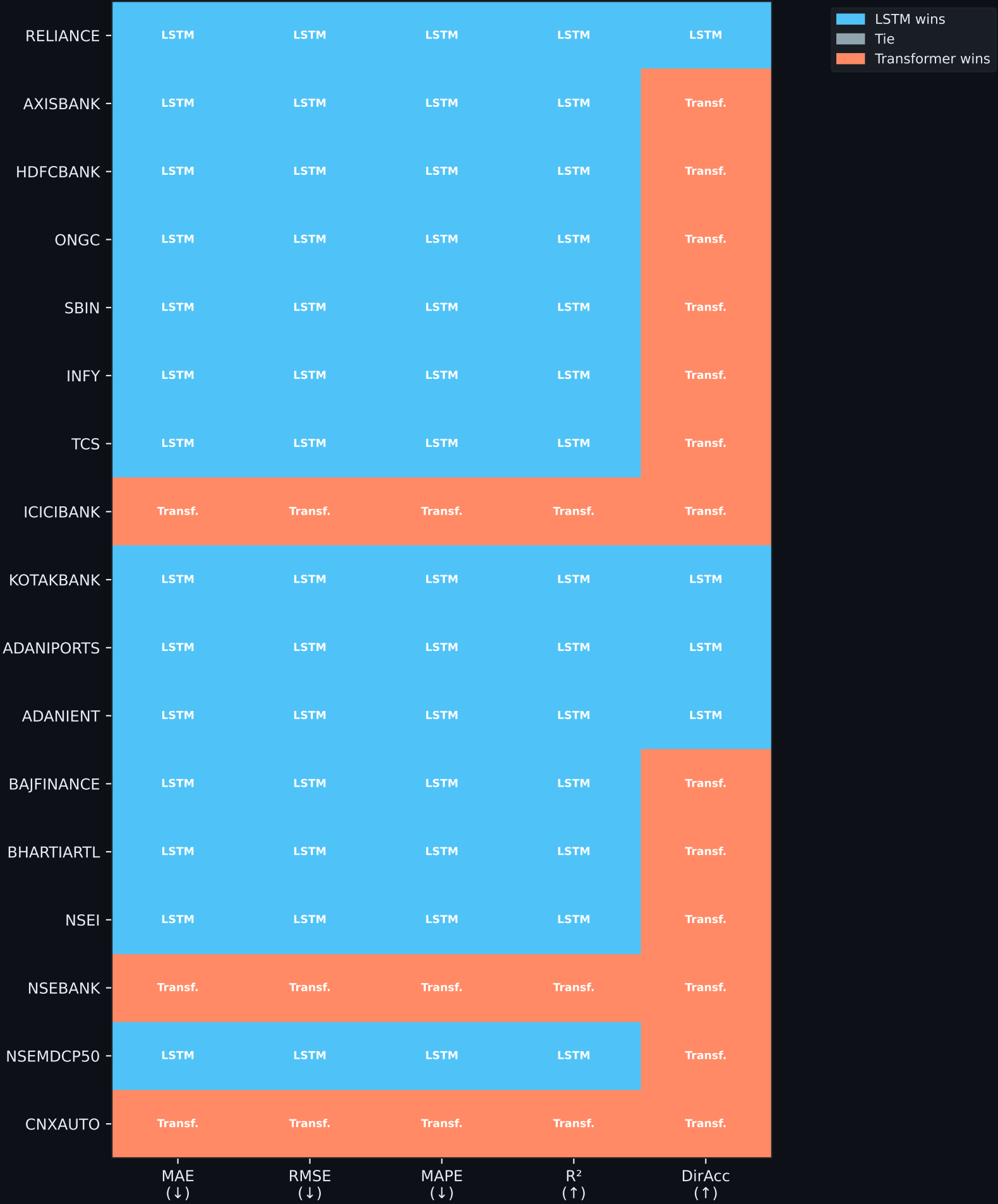
Model Comparison — R²



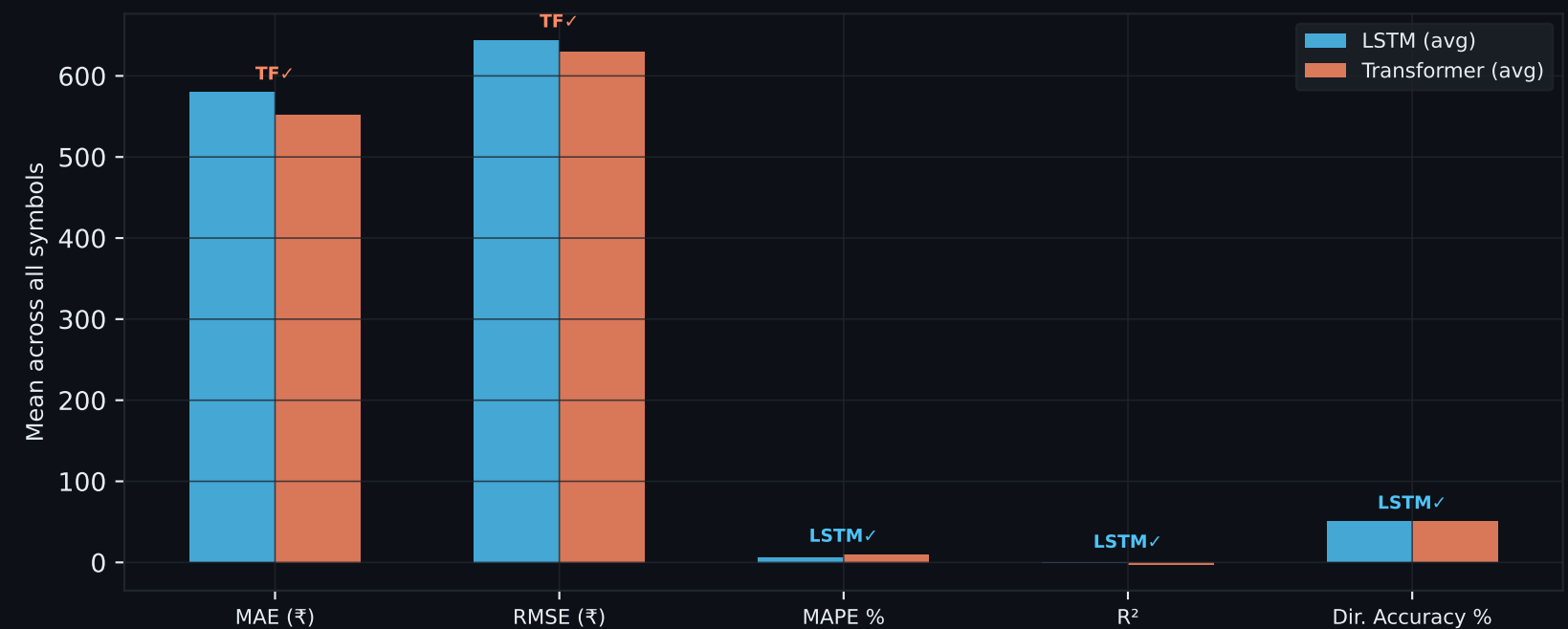
Model Comparison — Dir. Accuracy %



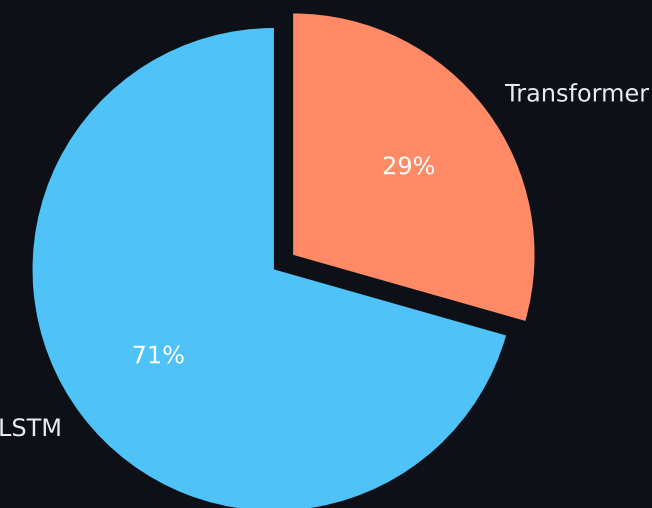
Win Heatmap — LSTM vs Transformer



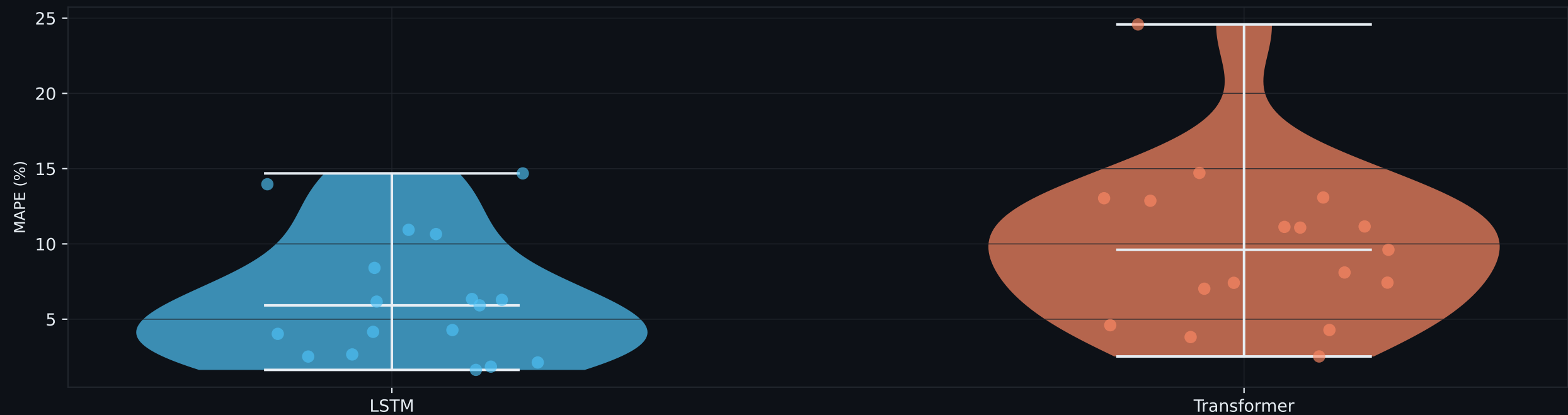
Aggregate Performance Summary



Win Share
(all metrics × symbols)

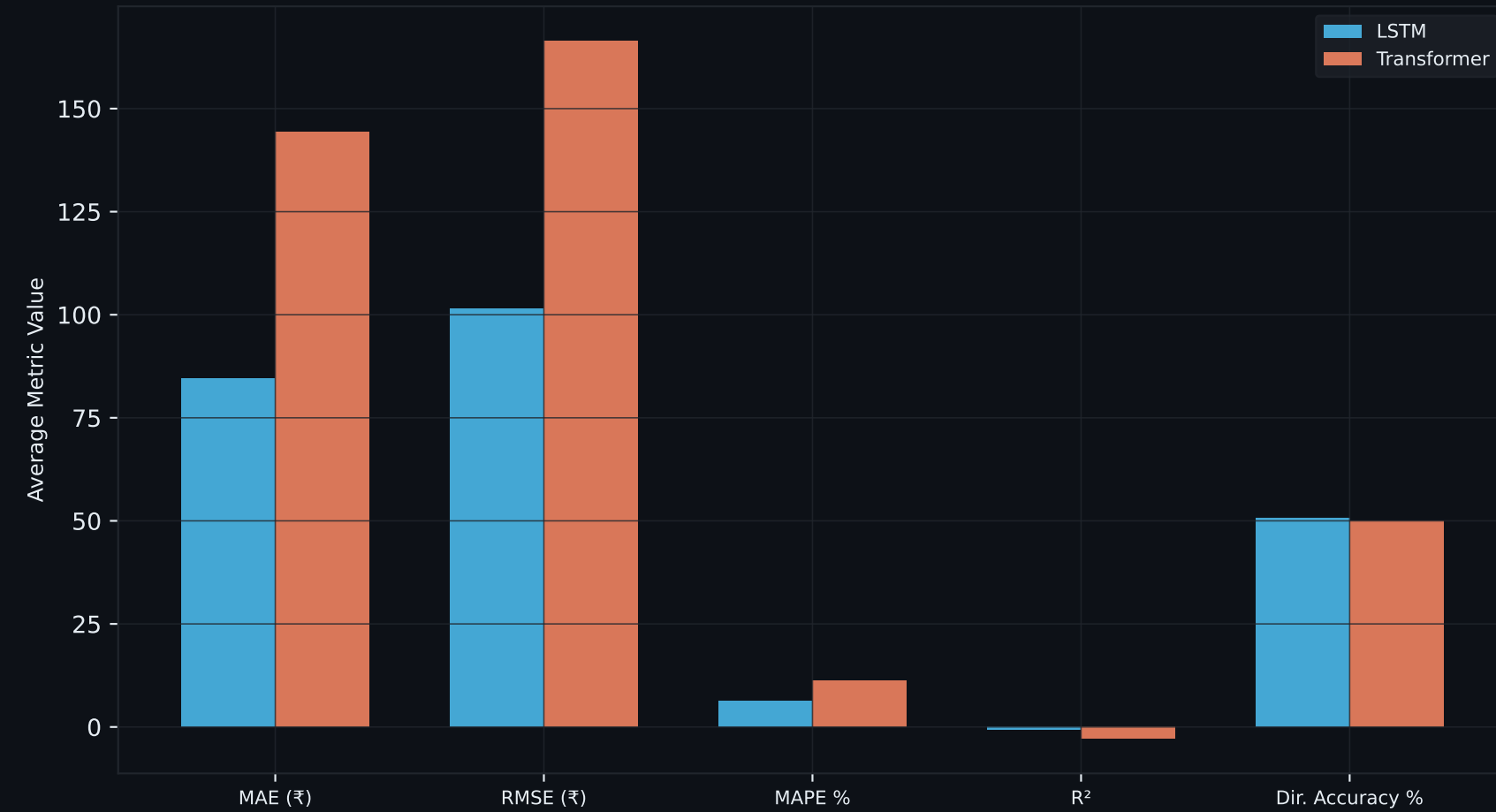


MAPE % Distribution (Test Split)

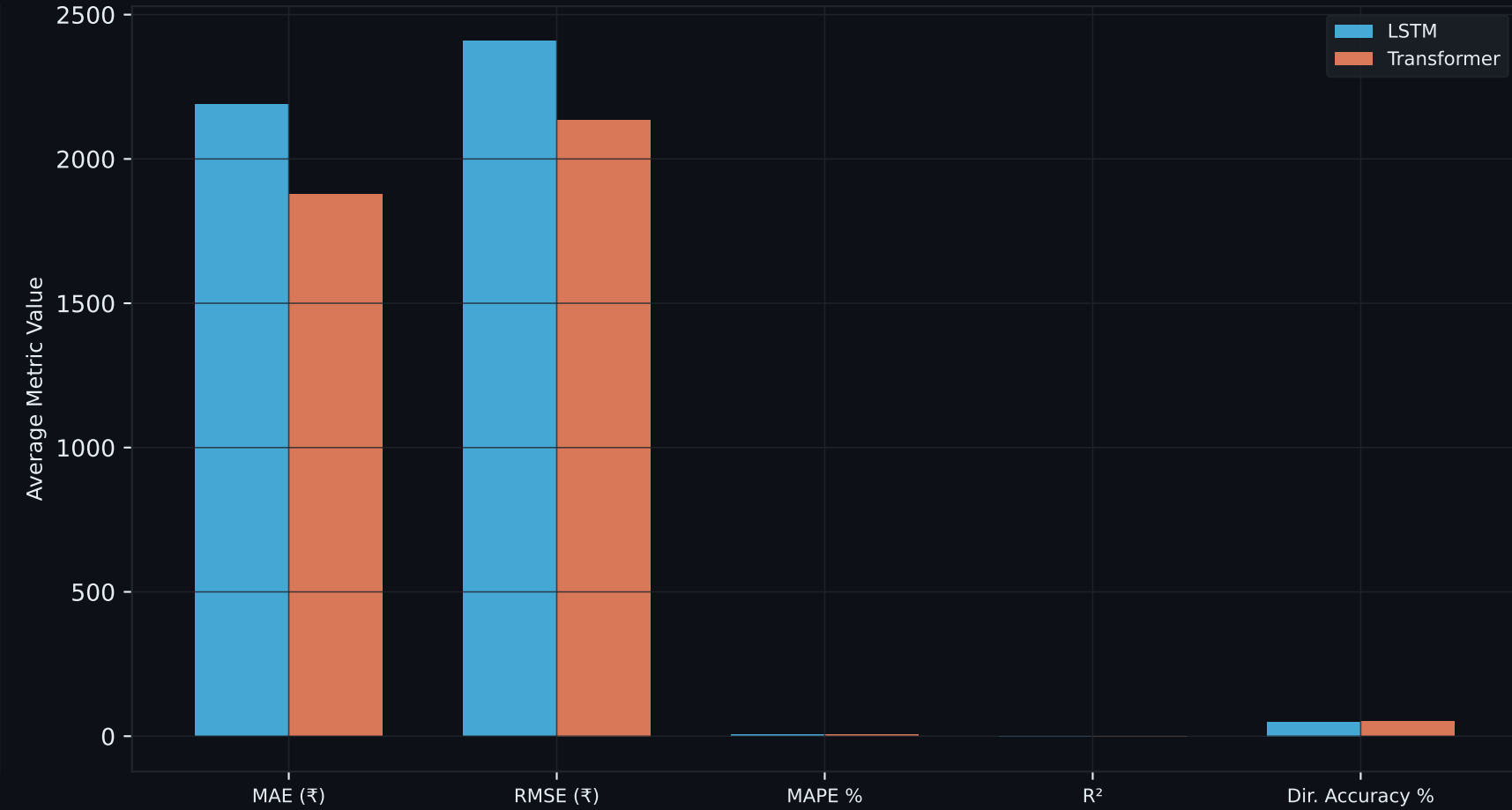


Average Metrics — Stocks vs Indices

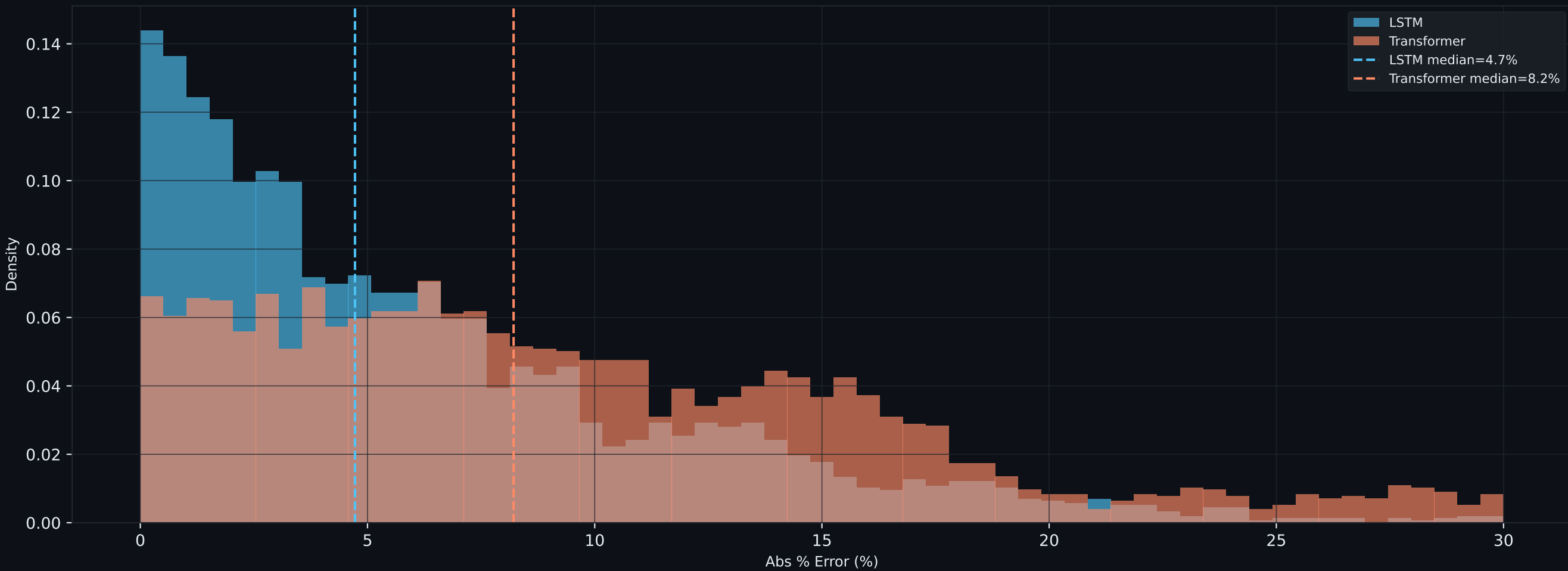
Stocks



Indices



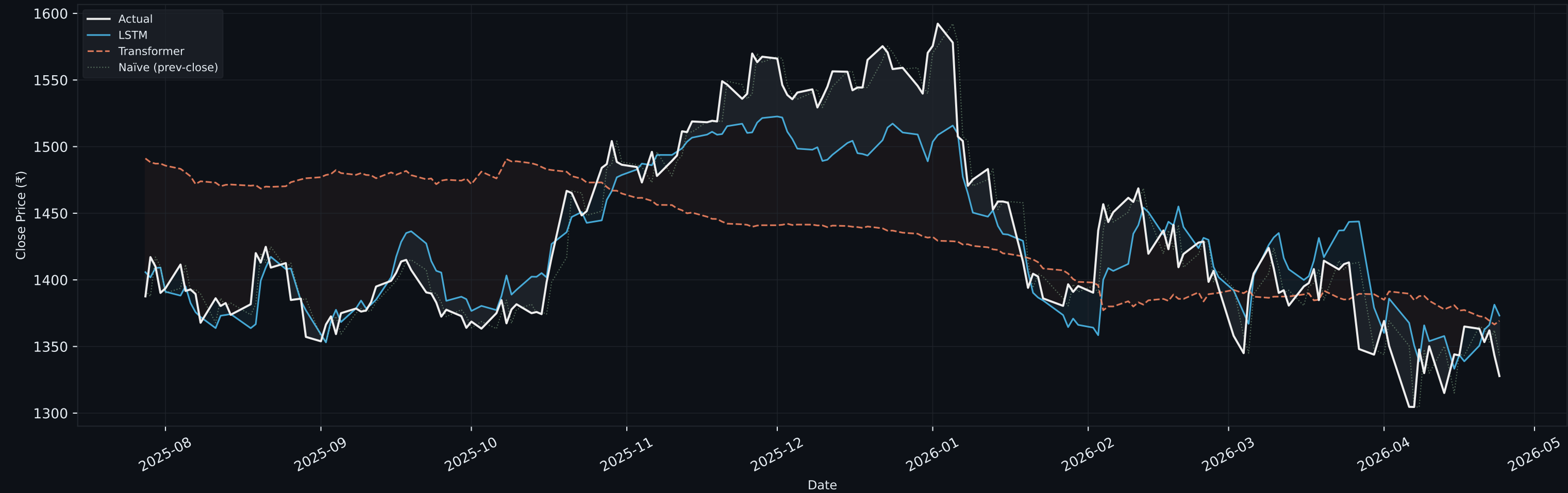
Absolute % Error Distribution Across All Symbols (Test Split)



Full Metrics Table — Fresh Inference (Test Split)

Symbol	MAE LSTM	MAE Transf.	RMSE LSTM	RMSE Transf.	MAPE LSTM	MAPE Transf.	R² LSTM	R² Transf.	DirAcc LSTM	DirAcc Transf.	Naive MAE
RELIANCE	23.7	66.4	30.7	77.5	1.63%	4.60%	0.812	-0.195	55.7%	49.7%	13.0
AXISBANK	53.8	102.5	69.0	119.4	4.16%	8.10%	0.511	-0.463	46.4%	49.2%	13.3
HDFCBANK	60.8	107.0	67.9	116.2	6.28%	11.13%	0.124	-1.563	55.2%	57.9%	7.7
ONGC	11.0	32.7	13.3	33.8	4.28%	13.08%	0.413	-2.796	48.1%	51.4%	2.4
SBIN	145.7	153.9	177.3	188.5	13.96%	14.71%	-1.295	-1.594	44.3%	45.4%	9.6
INFY	30.2	141.0	40.9	168.2	2.13%	9.61%	0.892	-0.831	48.6%	49.2%	17.4
TCS	167.3	300.7	251.2	405.3	6.34%	11.16%	0.134	-1.253	44.3%	47.0%	30.8
ICICIBANK	82.4	59.1	89.7	70.5	5.92%	4.28%	-1.817	-0.737	59.6%	59.6%	11.8
KOTAKBANK	16.9	53.7	20.3	58.1	4.03%	12.87%	0.151	-5.974	57.9%	53.0%	3.9
ADANIPORTS	90.8	108.6	103.0	125.3	6.17%	7.42%	-1.007	-1.974	51.9%	49.7%	17.8
ADANIENT	58.7	288.5	71.6	307.7	2.66%	13.04%	0.861	-1.558	51.9%	43.2%	33.4
BAJFINANCE	145.2	240.4	157.1	248.9	14.69%	24.58%	-4.517	-12.852	46.4%	46.4%	12.7
BHARTIARTL	213.3	222.7	227.6	243.4	10.65%	11.08%	-4.002	-4.721	49.7%	49.7%	17.2
NSEI	464.4	633.8	557.1	773.0	1.85%	2.52%	0.606	0.241	50.3%	53.0%	155.2
NSEBANK	4900.2	4335.6	5356.7	4820.7	8.41%	7.43%	-3.803	-2.890	46.4%	47.0%	394.5
NSEMDCP50	422.3	643.7	501.7	759.6	2.52%	3.81%	0.249	-0.720	49.5%	53.8%	136.7
CNXAUTO	2968.9	1896.2	3217.7	2187.7	10.94%	7.02%	-4.681	-1.626	52.5%	52.5%	255.9

Test-Split Predictions — RELIANCE.NS



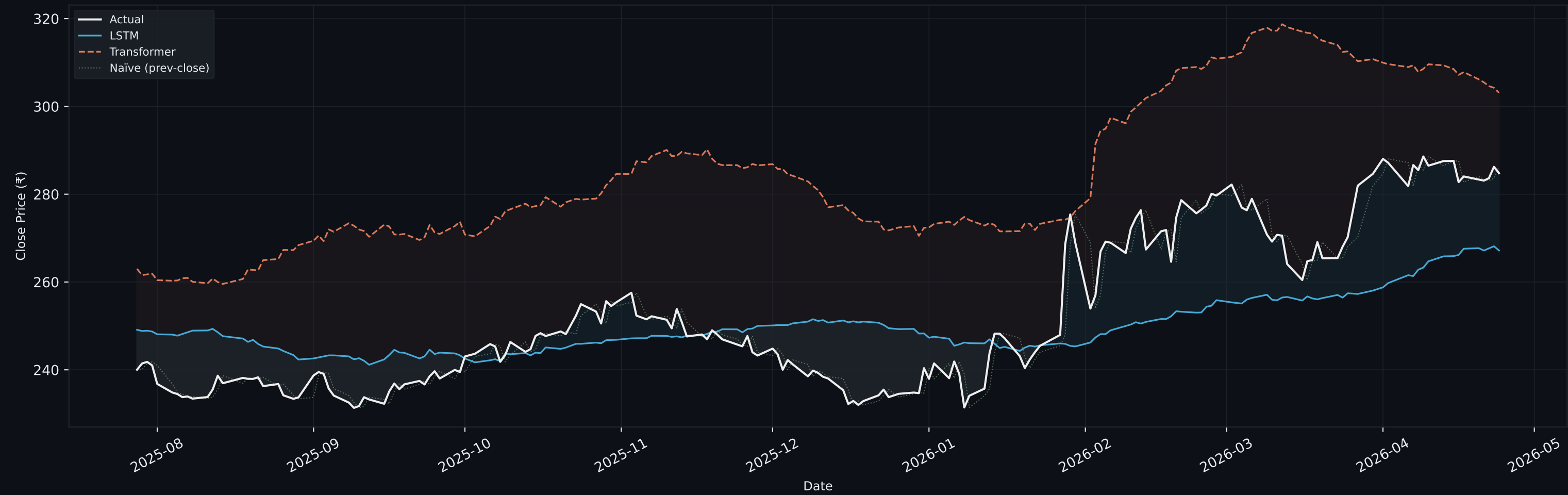
Test-Split Predictions — AXISBANK.NS



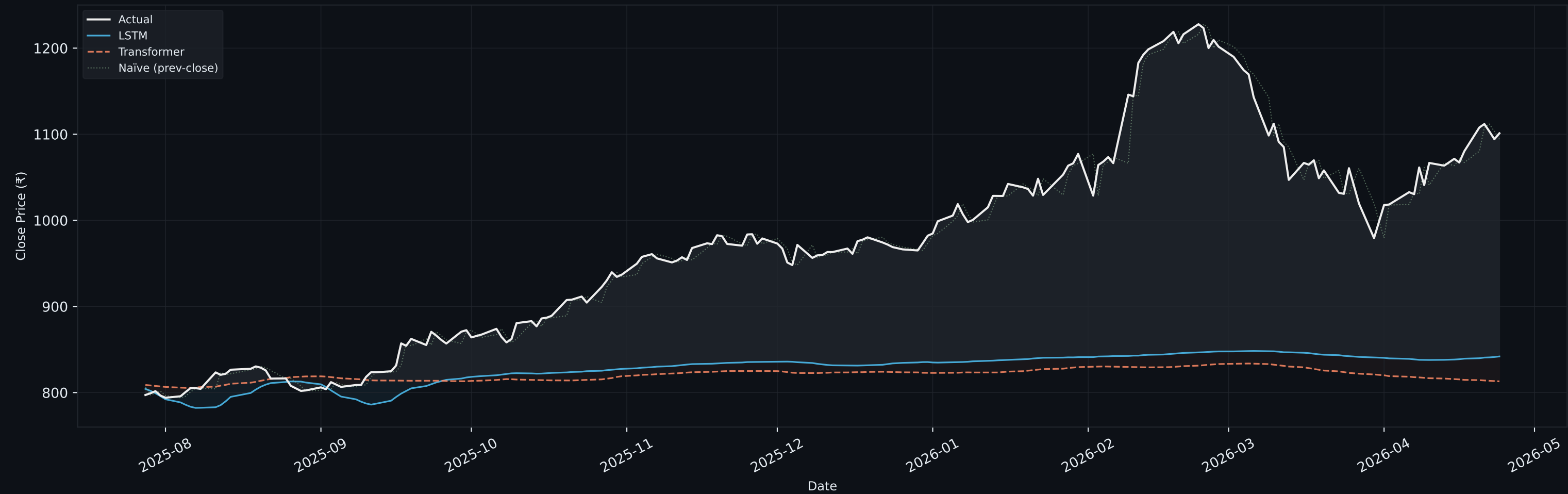
Test-Split Predictions — HDFCBANK.NS



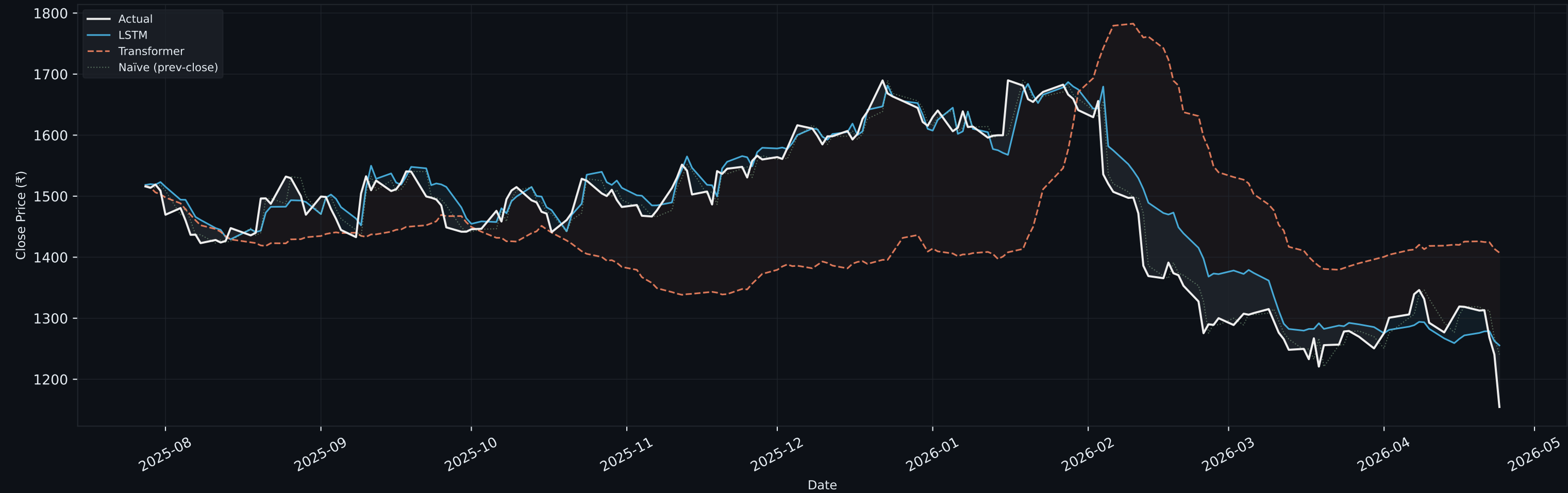
Test-Split Predictions — ONGC.NS



Test-Split Predictions — SBIN.NS



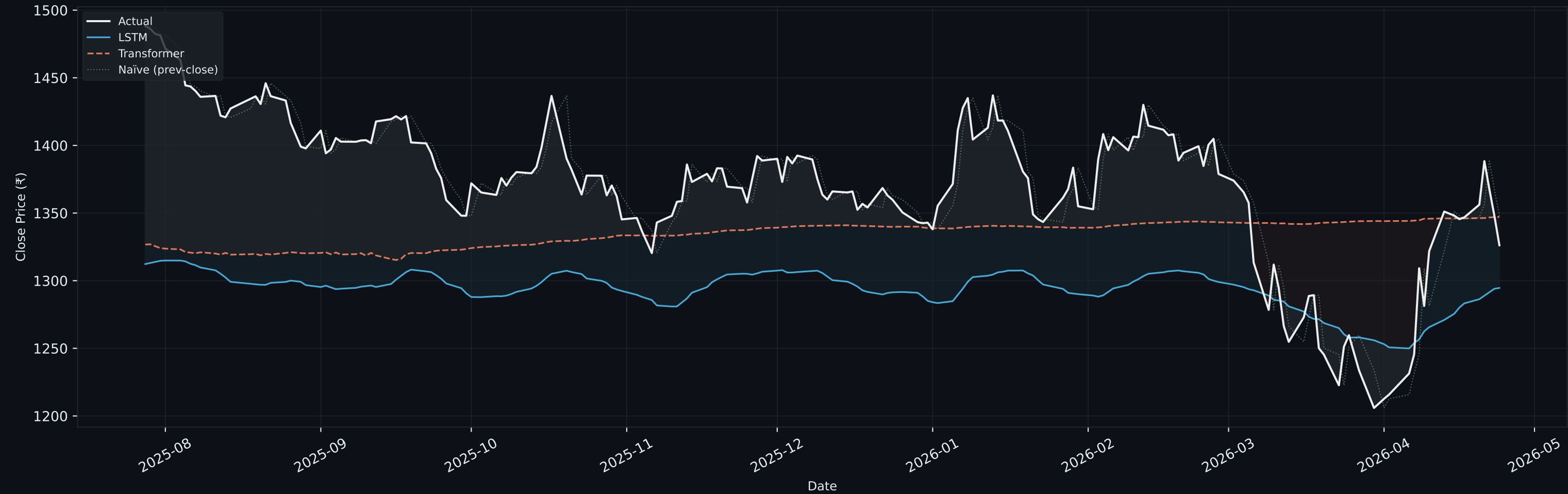
Test-Split Predictions — INFY.NS



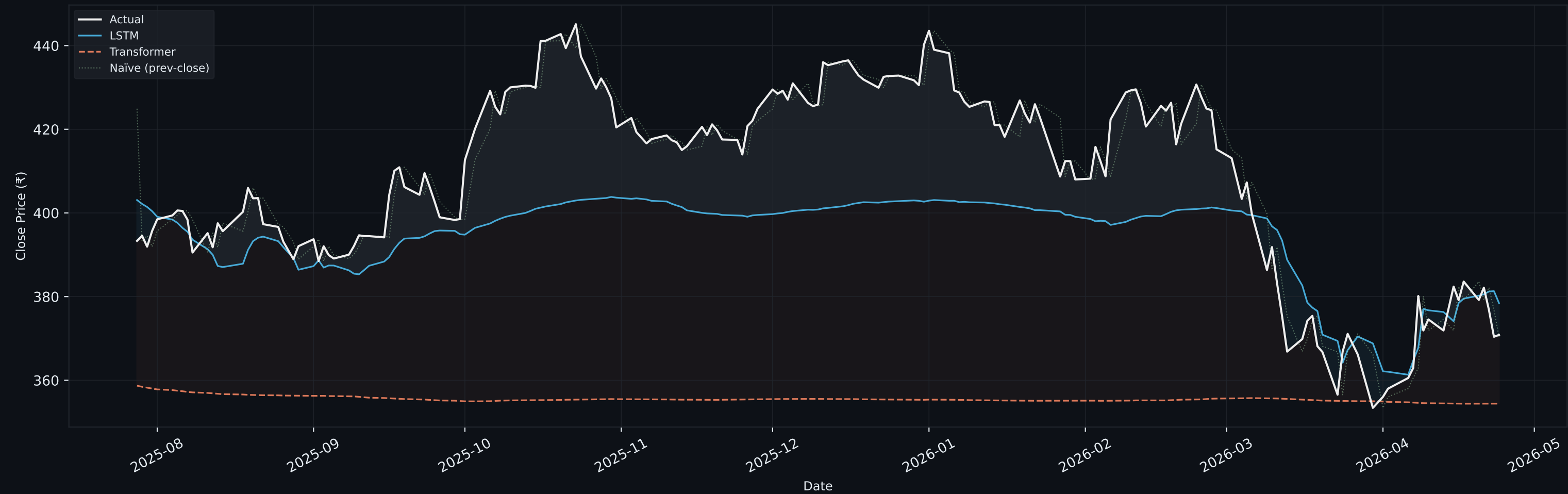
Test-Split Predictions — TCS.NS



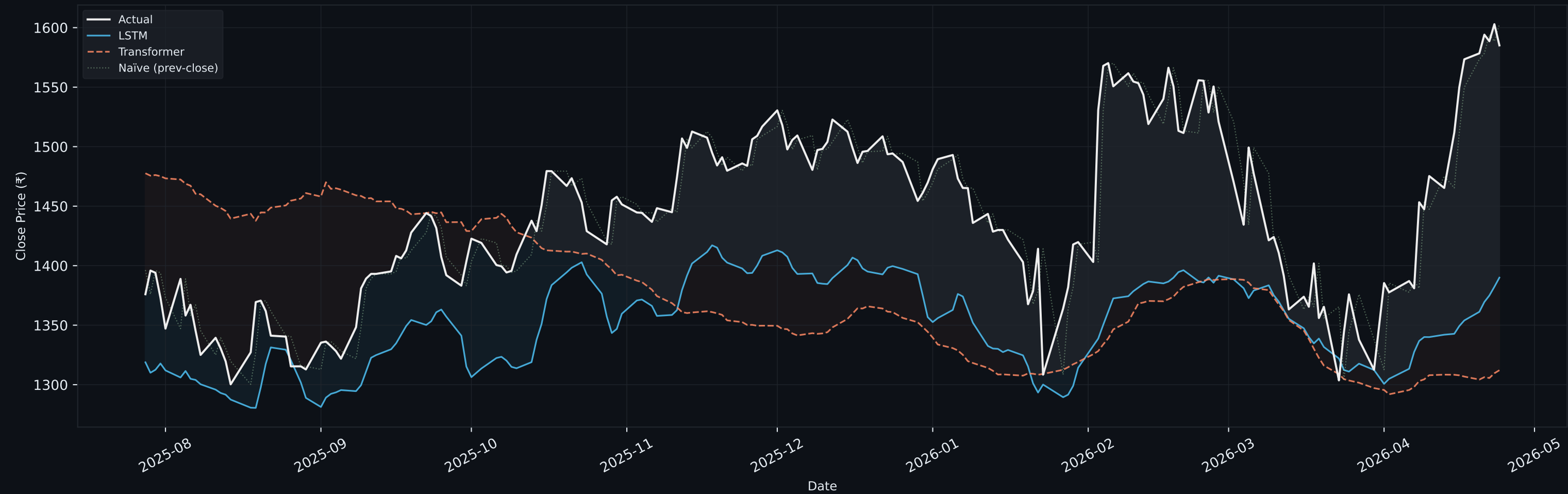
Test-Split Predictions — ICICIBANK.NS



Test-Split Predictions — KOTAKBANK.NS



Test-Split Predictions — ADANI PORTS.NS



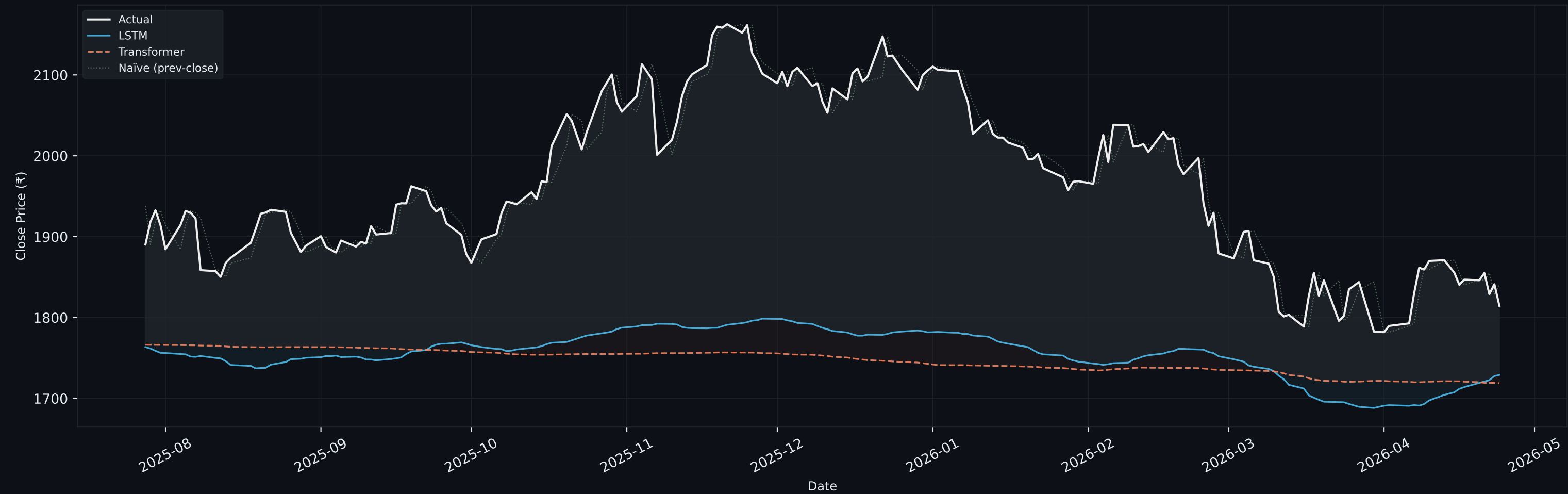
Test-Split Predictions — ADANIENT.NS



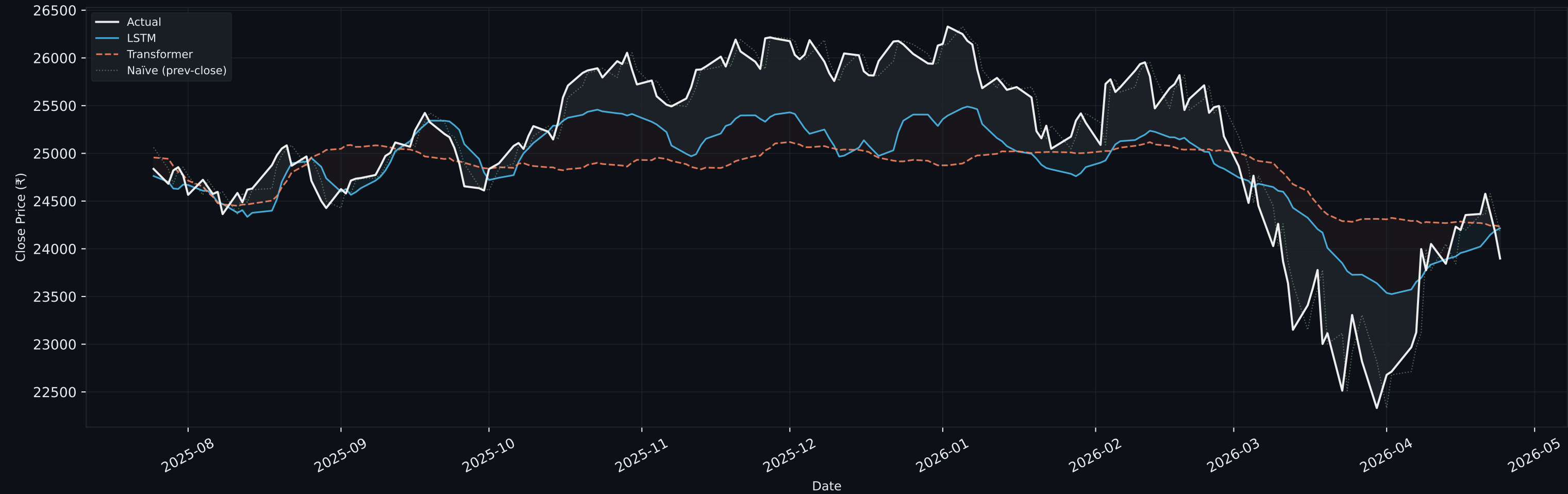
Test-Split Predictions — BAJFINANCE.NS



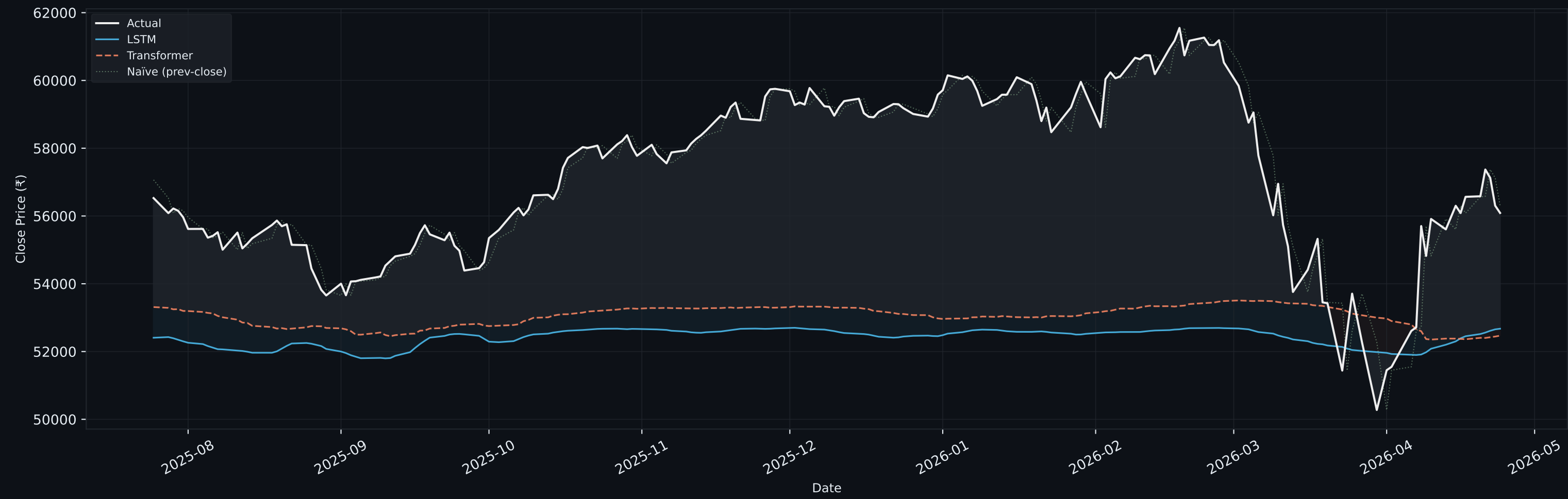
Test-Split Predictions — BHARTIARTL.NS



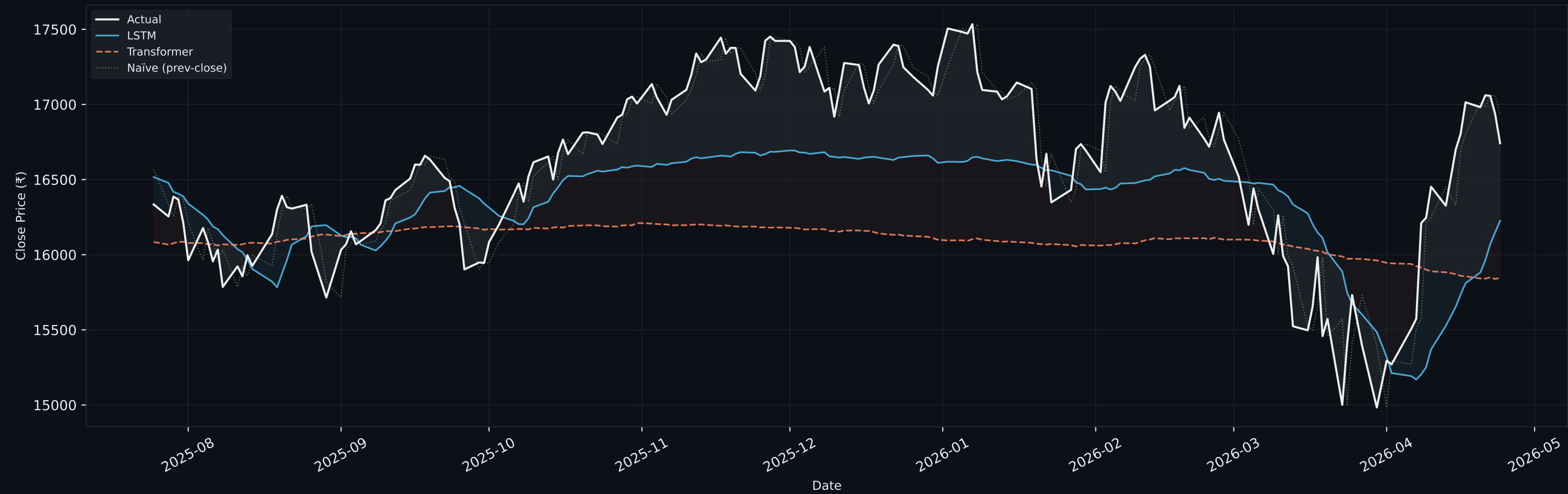
Test-Split Predictions — $\hat{NSEI}$



Test-Split Predictions — ^NSEBANK



Test-Split Predictions — $\hat{NSEMDCP50}$



Test-Split Predictions — ^CNXAUTO

